STM32 CubeMX

1. Description

1.1. Project

Project Name	Moppy
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	03/23/2026

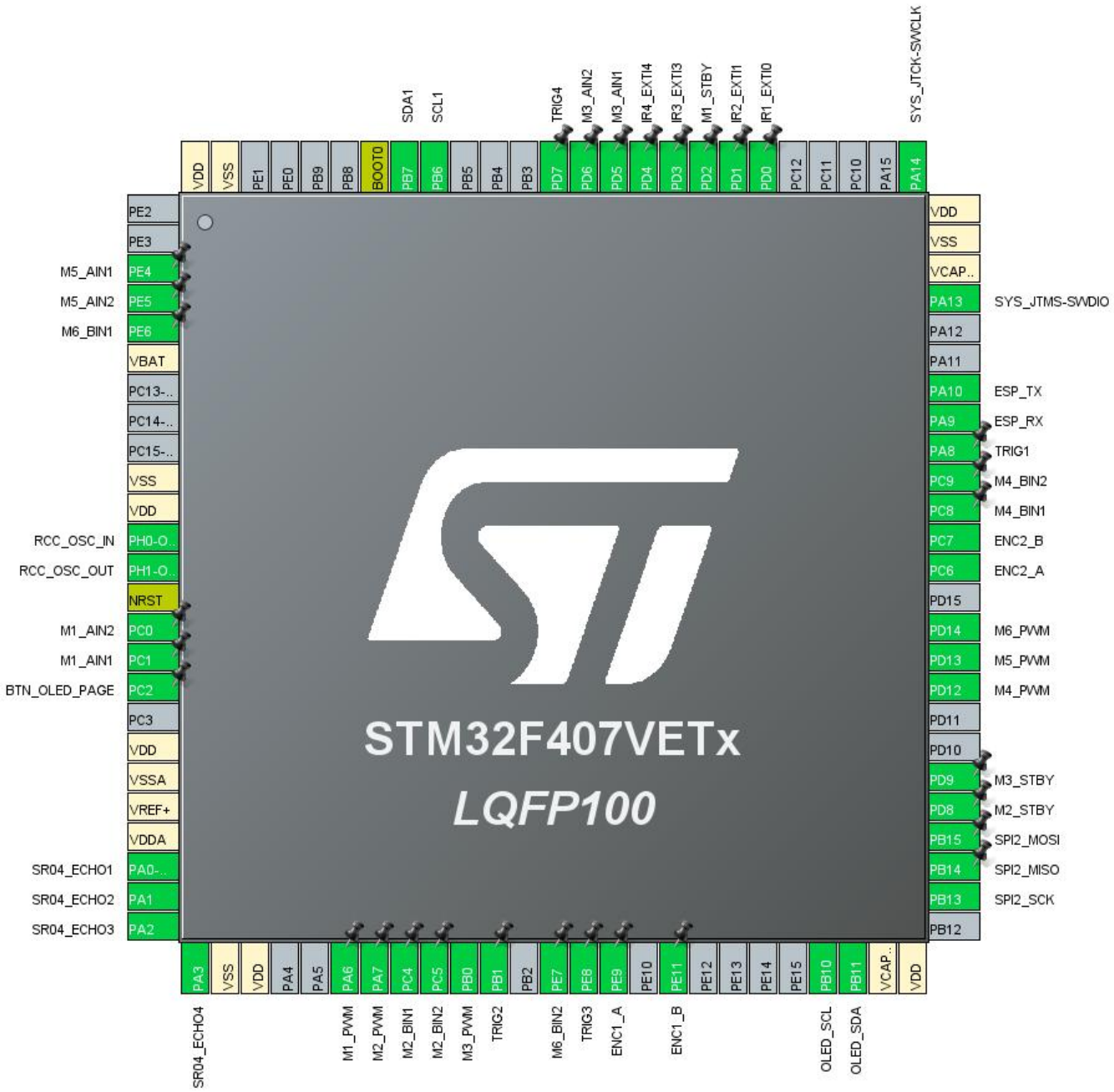
1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F407/417
MCU name	STM32F407VETx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	Arm Cortex-M4

2. Pinout Configuration



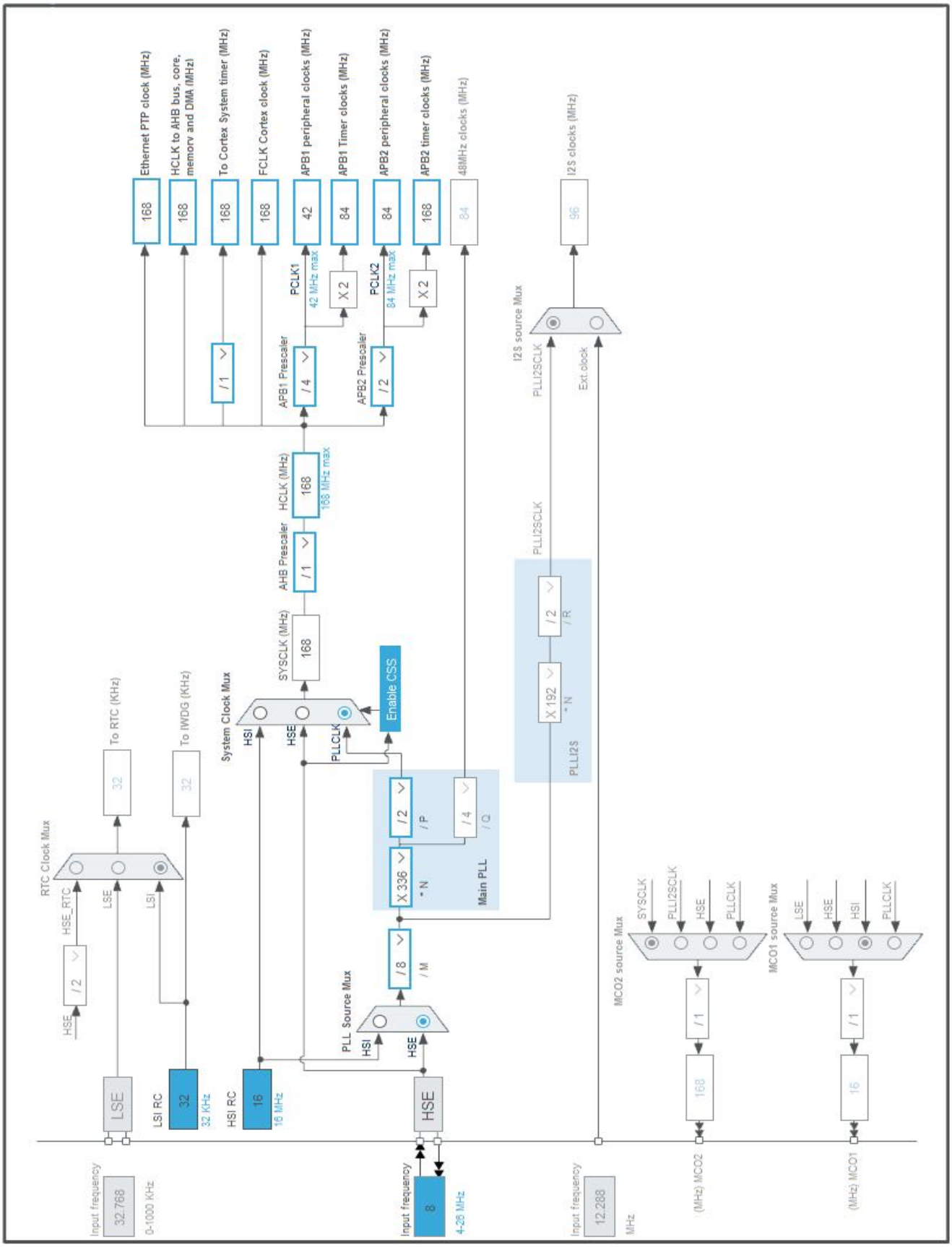
3. Pins Configuration

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
3	PE4 *	I/O	GPIO_Output	M5_AIN1
4	PE5 *	I/O	GPIO_Output	M5_AIN2
5	PE6 *	I/O	GPIO_Output	M6_BIN1
6	VBAT	Power		
10	VSS	Power		
11	VDD	Power		
12	PH0-OSC_IN	I/O	RCC_OSC_IN	
13	PH1-OSC_OUT	I/O	RCC_OSC_OUT	
14	NRST	Reset		
15	PC0 *	I/O	GPIO_Output	M1_AIN2
16	PC1 *	I/O	GPIO_Output	M1_AIN1
17	PC2 *	I/O	GPIO_Input	BTN_OLED_PAGE
19	VDD	Power		
20	VSSA	Power		
21	VREF+	Power		
22	VDDA	Power		
23	PA0-WKUP	I/O	TIM2_CH1	SR04_ECHO1
24	PA1	I/O	TIM2_CH2	SR04_ECHO2
25	PA2	I/O	TIM2_CH3	SR04_ECHO3
26	PA3	I/O	TIM2_CH4	SR04_ECHO4
27	VSS	Power		
28	VDD	Power		
31	PA6	I/O	TIM3_CH1	M1_PWM
32	PA7	I/O	TIM3_CH2	M2_PWM
33	PC4 *	I/O	GPIO_Output	M2_BIN1
34	PC5 *	I/O	GPIO_Output	M2_BIN2
35	PB0	I/O	TIM3_CH3	M3_PWM
36	PB1 *	I/O	GPIO_Output	TRIG2
38	PE7 *	I/O	GPIO_Output	M6_BIN2
39	PE8 *	I/O	GPIO_Output	TRIG3
40	PE9	I/O	TIM1_CH1	ENC1_A
42	PE11	I/O	TIM1_CH2	ENC1_B
47	PB10	I/O	I2C2_SCL	OLED_SCL
48	PB11	I/O	I2C2_SDA	OLED_SDA
49	VCAP_1	Power		
50	VDD	Power		

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
52	PB13	I/O	SPI2_SCK	
53	PB14	I/O	SPI2_MISO	
54	PB15	I/O	SPI2_MOSI	
55	PD8 *	I/O	GPIO_Output	M2_STBY
56	PD9 *	I/O	GPIO_Output	M3_STBY
59	PD12	I/O	TIM4_CH1	M4_PWM
60	PD13	I/O	TIM4_CH2	M5_PWM
61	PD14	I/O	TIM4_CH3	M6_PWM
63	PC6	I/O	TIM8_CH1	ENC2_A
64	PC7	I/O	TIM8_CH2	ENC2_B
65	PC8 *	I/O	GPIO_Output	M4_BIN1
66	PC9 *	I/O	GPIO_Output	M4_BIN2
67	PA8 *	I/O	GPIO_Output	TRIG1
68	PA9	I/O	USART1_TX	ESP_RX
69	PA10	I/O	USART1_RX	ESP_TX
72	PA13	I/O	SYS_JTMS-SWDIO	
73	VCAP_2	Power		
74	VSS	Power		
75	VDD	Power		
76	PA14	I/O	SYS_JTCK-SWCLK	
81	PD0	I/O	GPIO_EXTI0	IR1_EXTI0
82	PD1	I/O	GPIO_EXTI1	IR2_EXTI1
83	PD2 *	I/O	GPIO_Output	M1_STBY
84	PD3	I/O	GPIO_EXTI3	IR3_EXTI3
85	PD4	I/O	GPIO_EXTI4	IR4_EXTI4
86	PD5 *	I/O	GPIO_Output	M3_AIN1
87	PD6 *	I/O	GPIO_Output	M3_AIN2
88	PD7 *	I/O	GPIO_Output	TRIG4
92	PB6	I/O	I2C1_SCL	SCL1
93	PB7	I/O	I2C1_SDA	SDA1
94	BOOT0	Boot		
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F4
Line	STM32F407/417
MCU	STM32F407VETx
Datasheet	DS8626_Rev8

1.2. Parameter Selection

Temperature	25
Vdd	3.3

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

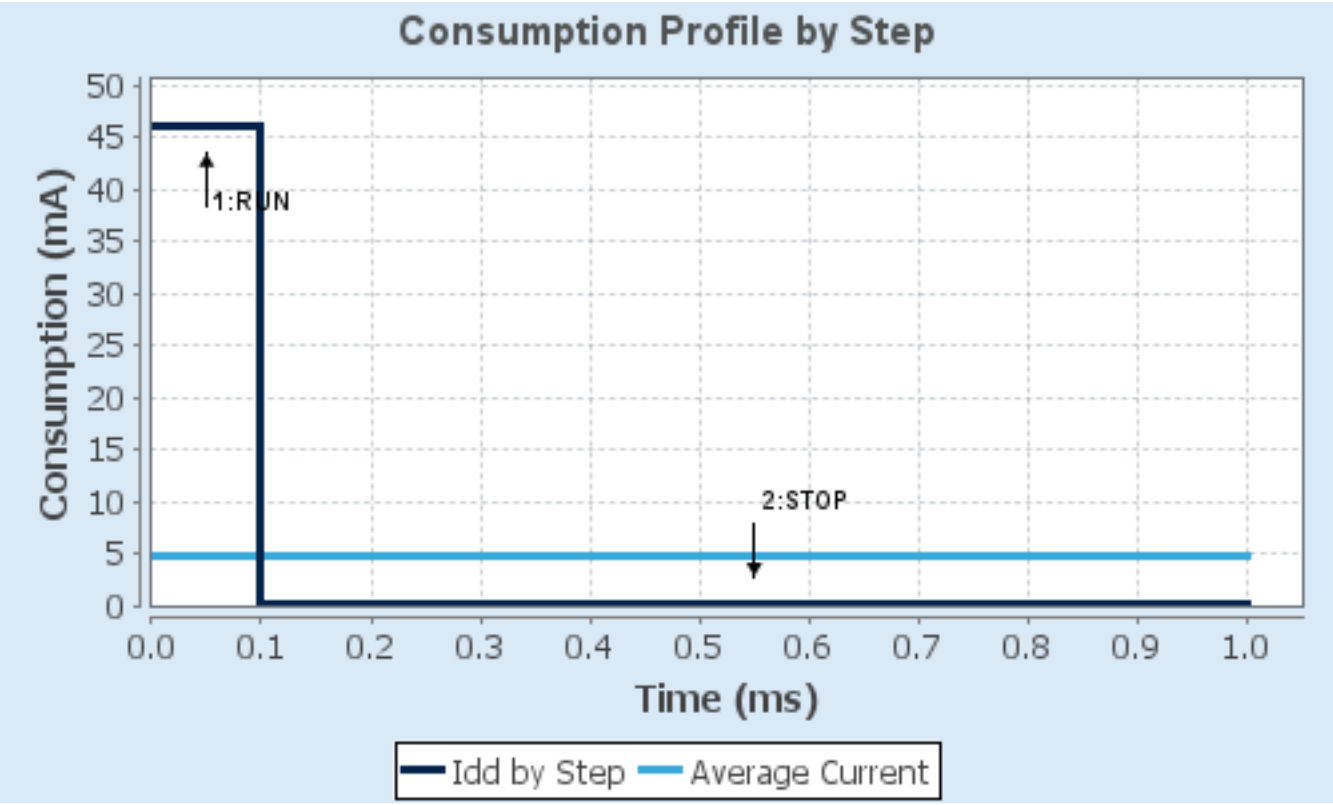
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.3	3.3
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	FLASH	n/a
CPU Frequency	168 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LP Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	46 mA	280 µA
Duration	0.1 ms	0.9 ms
DMIPS	210.0	0.0
Ta Max	98.47	104.96
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	4.85 mA
Battery Life	29 days, 4 hours	Average DMIPS	210.0 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
50
45
40
35
30
25
20
15
10
5
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	Moppy
Project Folder	C:\Users\DMXHP\Documents\Programing_Files_IoT\STM32CubeIDE\workspace
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_F4 V1.28.3
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_I2C1_Init	I2C1
5	MX_I2C2_Init	I2C2
6	MX_TIM2_Init	TIM2
7	MX_TIM3_Init	TIM3
8	MX_TIM4_Init	TIM4
9	MX_TIM6_Init	TIM6
10	MX_USART1_UART_Init	USART1
11	MX_TIM1_Init	TIM1

Rank	Function Name	Peripheral Instance Name
12	MX_TIM8_Init	TIM8
13	MX_SPI2_Init	SPI2

3. Peripherals and Middlewares Configuration

3.1. I2C1

I2C: I2C

3.1.1. Parameter Settings:

Master Features:

I2C Speed Mode	**Fast Mode ***
I2C Clock Speed (Hz)	400000
Fast Mode Duty Cycle	Duty cycle Tlow/Thigh = 2

Slave Features:

Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

3.2. I2C2

I2C: I2C

3.2.1. Parameter Settings:

Master Features:

I2C Speed Mode	**Fast Mode ***
I2C Clock Speed (Hz)	400000
Fast Mode Duty Cycle	Duty cycle Tlow/Thigh = 2

Slave Features:

Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

3.3. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.3.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	5 WS (6 CPU cycle)
RCC Parameters:	
HSI Calibration Value	16
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

3.4. SPI2

Mode: Full-Duplex Master

3.4.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	**21.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSS Signal Type	Software

3.5. SYS

Debug: Serial Wire

Timebase Source: SysTick

3.6. TIM1

Combined Channels: Encoder Mode

3.6.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.7. TIM2

Clock Source : Internal Clock

Channel1: Input Capture direct mode

Channel2: Input Capture direct mode

Channel3: Input Capture direct mode

Channel4: Input Capture direct mode

3.7.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**84 - 1 ***
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Input Capture Channel 1:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

Input Capture Channel 2:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

Input Capture Channel 3:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

Input Capture Channel 4:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

3.8. TIM3

Clock Source : Internal Clock

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

Channel3: PWM Generation CH3

3.8.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**3 ***
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	**999 ***
Internal Clock Division (CKD)	No Division
auto-reload preload	**Enable ***

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)

Trigger Event Selection	Reset (UG bit from TIMx_EGR)

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.9. TIM4

Clock Source : Internal Clock

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

Channel3: PWM Generation CH3

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**3 ***
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	**999 ***
Internal Clock Division (CKD)	No Division
auto-reload preload	**Enable ***

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

PWM Generation Channel 1:

Mode	PWM mode 1

Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.10. TIM6

mode: Activated

3.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**8399 ***
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	**9 ***
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	**Update Event ***

3.11. TIM8

Combined Channels: Encoder Mode

3.11.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535

Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.12. USART1

Mode: Asynchronous

3.12.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
I2C1	PB6	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	**Very High ***	SCL1
	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	**Very High ***	SDA1
I2C2	PB10	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	**Very High ***	OLED_SCL
	PB11	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	**Very High ***	OLED_SDA
RCC	PH0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SPI2	PB13	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
TIM1	PE9	TIM1_CH1	Alternate Function Push Pull	**Pull-up ***	Low	ENC1_A
	PE11	TIM1_CH2	Alternate Function Push Pull	**Pull-up ***	Low	ENC1_B
TIM2	PA0-WKUP	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	SR04_ECHO1
	PA1	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	SR04_ECHO2
	PA2	TIM2_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	SR04_ECHO3
	PA3	TIM2_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	SR04_ECHO4
TIM3	PA6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	M1_PWM
	PA7	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	M2_PWM
	PB0	TIM3_CH3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	M3_PWM
TIM4	PD12	TIM4_CH1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	M4_PWM
	PD13	TIM4_CH2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High**	M5_PWM

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					*	
	PD14	TIM4_CH3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	M6_PWM
TIM8	PC6	TIM8_CH1	Alternate Function Push Pull	**Pull-up ***	Low	ENC2_A
	PC7	TIM8_CH2	Alternate Function Push Pull	**Pull-up ***	Low	ENC2_B
USART1	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	ESP_RX
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	ESP_TX
GPIO	PE4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M5_AIN1
	PE5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M5_AIN2
	PE6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M6_BIN1
	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_AIN2
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_AIN1
	PC2	GPIO_Input	Input mode	**Pull-up ***	n/a	BTN_OLED_PAGE
	PC4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_BIN1
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_BIN2
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRIG2
	PE7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M6_BIN2
	PE8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRIG3
	PD8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_STBY
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M3_STBY
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M4_BIN1
	PC9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M4_BIN2
	PA8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRIG1
	PD0	GPIO_EXTI0	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	IR1_EXTI0
	PD1	GPIO_EXTI1	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	IR2_EXTI1
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_STBY
	PD3	GPIO_EXTI3	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	IR3_EXTI3
	PD4	GPIO_EXTI4	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	IR4_EXTI4
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M3_AIN1
	PD6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M3_AIN2
	PD7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRIG4

4.2. DMA configuration

DMA request	Stream	Direction	Priority
USART1_RX	DMA2_Stream2	Peripheral To Memory	**High ***
USART1_TX	DMA2_Stream7	Memory To Peripheral	Low

USART1_RX: DMA2_Stream2 DMA request Settings:

Mode:	**Circular ***
Use fifo:	Disable
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

USART1_TX: DMA2_Stream7 DMA request Settings:

Mode:	Normal
Use fifo:	Disable
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
TIM2 global interrupt	true	0	0
USART1 global interrupt	true	0	0
TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	true	0	0
DMA2 stream2 global interrupt	true	0	0
DMA2 stream7 global interrupt	true	0	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
EXTI line0 interrupt	unused		
EXTI line1 interrupt	unused		
EXTI line3 interrupt	unused		
EXTI line4 interrupt	unused		
TIM1 break interrupt and TIM9 global interrupt	unused		
TIM1 update interrupt and TIM10 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM11 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM3 global interrupt	unused		
TIM4 global interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
SPI2 global interrupt	unused		
TIM8 break interrupt and TIM12 global interrupt	unused		
TIM8 update interrupt and TIM13 global interrupt	unused		
TIM8 trigger and commutation interrupts and	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
TIM14 global interrupt			
TIM8 capture compare interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
TIM2 global interrupt	false	true	true
USART1 global interrupt	false	true	true
TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	false	true	true
DMA2 stream2 global interrupt	false	true	true
DMA2 stream7 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f405-415_407-417_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f405-415_407-417_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f4-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
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